Open Knowledge for a Sustainable Future: Research, Ethics, and Wikipedia

Week 5 — FAIR: making data findable and reusable

Francis Bond (**Academic**) Pavel Bednařík (Wiki)

Palacký University Olomouc | Wikimedia ČR

Today

1. FAIR Data: Findable, Accessible, Interoperable, Reusable
2. Open Licenses and Publications
3. Practice!

Roadmap

1. FAIR Data: Findable, Accessible, Interoperable, Reusable
   - Findable
   - Accessible
   - Interoperable
   - Reusable
2. Open Licenses and Publications
3. Practice!

Why FAIR Data?

- **Findable** — So that other researchers (and future you) can actually discover your data.
  If no one can find it, it might as well not exist.
- **Accessible** — So that once found, data can be retrieved easily and safely.
  Access should be possible even years later, with clear rules if restrictions apply.
- **Interoperable** — So that data from different projects can be combined or compared.
  Shared formats and vocabularies let computers and people understand each other.
- **Reusable** — So that data can be meaningfully used beyond its original purpose.
  Good documentation and clear licensing allow others to build on your work.

FAIR data makes research more transparent, verifiable, and sustainable.
Sources: (Wilkinson et al., 2016), FAIR Data 101 (ARDC, CC BY 4.0; accessed 2026-09-20)

Findable: why it matters

- Data has little value if no one can locate it.
- Clear titles, metadata, and persistent identifiers (like DOIs) make your work visible to both people and search engines.
- When data is easy to find, it is easier to cite — increasing your impact and ensuring credit.

Findable is putting your data on the map.

Making data findable

- Use descriptive, consistent names for datasets and files.
- Add rich metadata — who, what, when, where, why.
  - Use standard terms, so people can find them easily
- Register your dataset in a trusted repository with a permanent identifier (e.g., Zenodo, Dataverse).
- Cross-link: paper $\leftrightarrow$ dataset $\leftrightarrow$ code.
- This applies to governments and companies as well as academia
  - Making data findable should be part of making the data!

Identifiers for people: ORCID

- A DOI identifies a work; an ORCID iD identifies the person who made it
  - 16 digits: Francis's is 0000-0003-4973-8068
  - Free, run by a non-profit; it stays with you when you change institution or name
- It solves the name problem from week 4
  - Nurril Hirfana binte Mohamed Noor, Nurril Hirfana, or N. H. Mohamed Noor?
  - Or the many F. Bonds
- Journals, funders and repositories ask for it when you submit, and put it in the metadata (as in the `.zenodo.json` two slides on)
- Wikidata links people to it too: property P496
- Registration at orcid.org is free

Metadata inside the file: this PDF

`pdfinfo -custom wk-05-fair.pdf` (or *File* → *Properties* in a PDF reader):

```
Title:      Open Knowledge for a Sustainable Future: Research, Ethics, and Wikipedia - Week 5 - FAIR:
↪  making data findable and reusable
Author:     Francis Bond (Academic), Pavel Bednařík (Wiki)
Subject:    Open Knowledge for a Sustainable Future: Research, Ethics, and Wikipedia (Palacký
↪  University Olomouc)
Keywords:   open knowledge, research ethics, Wikipedia, FAIR data, CARE principles, academic writing,
↪  peer review
License:    CC BY 4.0
LicenseURL: https://creativecommons.org/licenses/by/4.0/
Producer:   LuaTeX-1.22.0
```

- A few lines in the LaTeX preamble; in Word: *File* → *Info* → *Properties*
- Search engines and reference managers read these fields
- The metadata travels with the file: email it, and the licence goes too

Metadata for a repository: `.zenodo.json`

```
{
  "title": "Open Knowledge for a Sustainable Future: Research, Ethics, and Wikipedia",
  "description": "Course materials (slides, handouts and website) for a course at Palacký University
  ↪  Olomouc on writing (both academic and for Wikipedia), finding, judging and citing sources, FAIR
  ↪  and CARE data, feedback and peer review.",
  "upload_type": "lesson",
  "creators": [
    {"name": "Bond, Francis", "affiliation": "Palacký University Olomouc",
     "orcid": "0000-0003-4973-8068"},
    {"name": "Bednařík, Pavel", "affiliation": "Wikimedia ČR"}
  ],
  "license": "cc-by-4.0",
  "language": "eng",
  "keywords": ["open knowledge", "research ethics", "Wikipedia", "FAIR data",
               "CARE principles", "academic writing", "peer review"],
  "related_identifiers": [
    {"identifier": "https://bond-lab.github.io/Open-Knowledge/",
     "relation": "isDocumentedBy", "resource_type": "other"}
  ]
}
```

Put this in a GitHub repository connected to Zenodo: each release is archived, with a DOI.

Accessible: why it matters

- Even well-described data is useless if people can't reach it.
- Accessibility ensures that others can actually download or view your work — now and in the future.
- It also clarifies boundaries: who can use the data, under what conditions.

Access is about clarity and stability, not about giving everything away. For sensitive language/cultural data, clarity of access rules is part of the ethics record.

Making data accessible

- Store your data in repositories that use open web standards (HTTP/HTTPS).
  - Use persistent repositories
  - Support persistent repositories
  - Data from companies is typically very perishable
- Keep metadata available even if files must be restricted.
- Explain how others can request access if necessary.
- Test from a new computer — can someone else follow your instructions?

Interoperable: why it matters

- Science advances when we can combine data from many sources.
- Interoperability allows your work to connect with others'
  — across languages, disciplines, and tools.
- Shared standards prevent "data silos" that only one group can use.

Interoperable data talks easily to other data.

Making data interoperable

- Use open, non-proprietary formats (CSV, JSON, XML).
  - Text is better than binary, as you can always read it
- Follow community conventions for terminology and encoding.
- Provide clear documentation of structure and meaning.
- Link to external vocabularies or identifiers when possible.
  - e.g. ISO 639 for the representation of languages and language groups.
    - English: en, eng (ISO 639-1, 2&3)
    - Czech: cs, cze/ces, ces (ISO 639-1, 2-B/T, 3)
    - the parts were consolidated into a single standard, ISO 639:2023, where they are now called sets — the codes are unchanged

Reusable: why it matters

- Good data should outlive the project that created it.
- Reuse lets others verify results, test new questions, and save time.
- Well-documented and licensed data reduce duplication and waste.
- Producing data is the job of many agencies, but if no one uses it, it is wasted effort, …

Reuse is the payoff — the reward for all your FAIR effort.

Making data reusable

- Include full context: how the data was collected, cleaned, and analysed.
- Choose a clear licence (e.g., CC BY or CC BY-SA).
- Describe known limitations or ethical constraints.
- Provide examples of how to cite or acknowledge your dataset.

Citing the Open Multilingual Wordnet

If you use these wordnets, please cite the original projects who created them, if you got value from this aggregation/normalization, please cite Bond and Paik (2012).

References

Francis Bond and Kyonghee Paik (2012) A survey of wordnets and their licenses In *Proceedings of the 6th Global WordNet Conference (GWC 2012)*. Matsue. 64–71

Roadmap

1. FAIR Data: Findable, Accessible, Interoperable, Reusable
2. Open Licenses and Publications
3. Practice!

Open licences and the Open Definition

- **Open data** means more than free access — it means legal permission to *use, modify, and share* without discrimination.
- The **Open Definition** (Open Knowledge Foundation) states that data is open if "anyone can freely access, use, modify, and share for any purpose" — subject only to requirements of attribution and share-alike.
- Common open licences:
  - **CC BY** – reuse with attribution.
  - **CC BY-SA** – reuse with attribution and same licence (share-alike).
  - **CC0 /Public Domain** – no restrictions.
  - **ODC-BY** – for databases; attribution only.
  - **ODbL** – for databases; attribution *and* share-alike.
- Choosing the right licence ensures that your data remains reusable and legally safe.

Openness is a design choice—licensing makes it possible, clarity makes it trustworthy.

When not to use open licences

- Openness is a virtue, but not a universal rule.
- Some data should *not* be released under open licences because openness could cause harm.
- Examples include:
  - Personal or medical information that identifies individuals.
  - Cultural or linguistic materials shared under community protocols.
  - Locations of endangered species or sacred sites.
  - Data collected without full, informed consent for public reuse.
- In such cases, use restricted or tiered access, or licences that reflect community agreements.

Responsible openness means balancing transparency with care, consent, and context.

Where to publish: routes to open access

- A licence says what readers are allowed to do. The **route** decides whether they can read the work at all.
- **Gold**: the journal publishes the final version openly, and charges the author an article processing charge (APC) — hundreds to thousands of euros, usually paid by a funder or institution.
- **Diamond**: open to readers and free to authors, funded by universities or scholarly societies. Much of linguistics works this way.
- **Green**: publish anywhere, then deposit the accepted manuscript in a repository — your university's, or a subject archive such as arXiv or LingBuzz.
- **Predatory venues** take the APC and skip the peer review. Check the journal against the DOAJ, and see whether you recognise anyone on the editorial board.

Roadmap

1. FAIR Data: Findable, Accessible, Interoperable, Reusable
2. Open Licenses and Publications
3. Practice!

Due today: your annotated bibliography

- **Five or more sources**, in APA 7, with **two or three sentences** on each
- Each annotation answers three questions:
  - What does this source *say* that bears on your question?
  - How far can you *trust* it — who wrote it, where did it appear, what is the evidence behind it?
  - What will you *use* it for: background, evidence, a position you argue against?
- It is not a summary exercise. An annotation that could have been written without reading the source is worth nothing to you in December, when you are writing the paper and cannot remember which one said what.
- Not graded on its own — it is the scaffolding for the paper, and we use it today.

Auditing sources for FAIR and CARE issues

- **30 minutes, in pairs.** Work from the annotated bibliography you brought today.
- **Part 1 (10 min): pitch your topic.** Tell your partner the question you want your paper to answer, then read them two of your annotations: your **best** source and your **shakiest**. Your partner asks: how do you know that one is any good?
- **Part 2 (15 min): audit one source each.** Pick the shaky one and check it:
  - **FAIR:** can you *find* it by DOI or stable URL, can you actually *read* it, is its data or method reusable, is the licence clear?
  - **CARE:** whose data or words does it rest on? Did they consent? Who benefits from it being published? (we will look at these in more detail next week)
- **Part 3 (5 min):** decide together — keep it, replace it, or cite it *with* its limitations stated. **Rewrite that annotation** before you leave.

Your first-draft paper: fitting the scientific style

- The upcoming assignment is your **first draft of a short academic paper.**
- Use the conventions of scientific writing:
  - Clear research question and motivation.
  - Evidence-based argument supported by credible sources.
  - Neutral tone; cautious interpretation; accurate citations.
- The work you have already done both helps:
  - The three-paragraph comparison of texts on water use in data centres → weighing sources against each other, and citing them in APA.
  - Wikipedia → writing neutrally and clearly for others.

I want to see how well you can use your sources to answer the question, as accurately as possible.

- In the peer-review session, you will read each other's papers and comment on them.

Acknowledgements I

- **FAIR material** draws on FAIR Data 101 (Australian Research Data Commons, CC BY 4.0) and Wilkinson et al. (2016).
- **AI disclosure: GPT-4o** in ChatGPT (OpenAI, 2025) was used to format the references and to draft an early version of these slides.
- **Claude Opus 5** (Anthropic, 2026) was used to check the slides: to resolve every DOI against Crossref, fetch every URL, and compare each quotation with its original. The prompt was of the form

```
Verify every claim, DOI, URL and quotation against a real source. Where you cannot
verify something, say so rather than guessing --- never invent a reference.
```

  Each correction names the source it was checked against. Responsibility for what remains is mine.

References I

- Anthropic. (2026). *Claude Opus 5 (Sep 20, 2026 version)* (Large language model). Retrieved September 20, 2026, from https://claude.ai/
- OpenAI. (2025). *ChatGPT (Mar 14, 2025 version)* (Large language model). Retrieved October 14, 2025, from https://chat.openai.com/
- Wilkinson, M. D., Dumontier, M., Aalbersberg, I. J., Appleton, G., Axton, M., Baak, A., et al. (2016). The fair guiding principles for scientific data management and stewardship. *Scientific Data*, *3*(160018). https://doi.org/10.1038/sdata.2016.18